IPROJECT POSTER · DIGITAL SYSTEMS

DIGITAL DOOR ACCESS SYSTEM USING COMBINATIONAL LOGIC

Validates identity and access inputs, then gives an immediate unlock or deny signal.

1 · INTRODUCTION / PROBLEM

FAST, LOW-COST ENTRY CONTROL

- Manual checks are inconsistent
- Fixed-input locks need clear ID, PIN, and alarm decision



2 · OUR SOLUTION

IMMEDIATE BINARY DECISION

- Combinational AND/OR/NOT controller
- Unlocks only when V=1, P=1, and A=0

3 · OBJECTIVES

- Verify ID and PIN inputs
- Control door unlock logic
- Trigger alarm status

4 · COMPONENTS REQUIRED

- Logic gates (AND, OR, NOT)
- Switches for ID, PIN, alarm
- LED indicators and breadboard

6 · CIRCUIT WORKING

- Inputs V, P, A feed gates
- Output U drives lock or LED

5 · BLOCK DIAGRAM

V (Valid ID) · P (Correct PIN) · A (Alarm) → AND/NOT logic → U (Door Unlock)

7 · TRUTH TABLE

V	0	0	0	0	1	1	1	1
P	0	0	1	1	0	0	1	1
A	0	1	0	1	0	1	0	1
U	0	0	0	0	0	0	**1 UNLOCK**	0

V = Valid ID · P = Correct PIN · A = Alarm · U = Door Unlock

8 · LOGIC DESIGN

$$U = V \wedge P \wedge \neg A$$

Unlock only when V=1, P=1, A=0.

9 · SIMULATION / IMPLEMENTATION

- Truth-table verification in simulation tool
- Breadboard build with LED output

10 · COURSE OUTCOMES

CO1 Digital Logic & Number System

CO2 Combinational Logic Design

CO3 Sequential Logic Design

CO4 Computer Arithmetic & Data Representation

CO5 Memory Organization & System Design

CO6 Hardware Description & Simulation

11 · ADVANTAGES

- Low cost, fast decision

12 · LIMITATIONS

- Fixed inputs, no memory

13 · APPLICATIONS / FUTURE

- Offices, labs; add sequential lock states

14 · CONCLUSION

- Simple AND/OR/NOT logic secures access

15 · TEAM MEMBERS

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